

The case $R = \mathbb{Z}$ is particularly simple. We can take S^m as the $(m+1)$ -skeleton of K_m , and similarly for K_n , so $K_m \wedge K_n$ has $S^m \wedge S^n$ as its $(m+n+1)$ -skeleton and we can obtain μ by extending the inclusion $S^m \wedge S^n = S^{m+n} \hookrightarrow K_{m+n}$.

It is not hard to prove the basic properties of cup product using this definition, and in particular the commutativity property becomes somewhat more transparent from this viewpoint. For example, when $R = \mathbb{Z}$, commutativity just comes down to the fact that the map $S^m \wedge S^n \rightarrow S^n \wedge S^m$ switching the factors has degree $(-1)^{mn}$ when regarded as a map of S^{m+n} .

Fibrations

Recall from §4.2 that a fibration is a map $p: E \rightarrow B$ having the homotopy lifting property with respect to all spaces. In a fiber bundle all the fibers are homeomorphic by definition, but this need not be true for fibrations. An example is the linear projection of a 2-simplex onto one of its edges, which is a fibration according to an exercise at the end of the section. The following result gives some evidence that fibrations should be thought of as a homotopy-theoretic analog of fiber bundles:

Proposition 4.61. *For a fibration $p: E \rightarrow B$, the fibers $F_b = p^{-1}(b)$ over each path component of B are all homotopy equivalent.*

Proof: A path $\gamma: I \rightarrow B$ gives rise to a homotopy $g_t: F_{\gamma(0)} \rightarrow B$ with $g_t(F_{\gamma(0)}) = \gamma(t)$. The inclusion $F_{\gamma(0)} \hookrightarrow E$ provides a lift $\tilde{g}_0$, so by the homotopy lifting property we have a homotopy $\tilde{g}_t: F_{\gamma(0)} \rightarrow E$ with $\tilde{g}_t(F_{\gamma(0)}) \subset F_{\gamma(t)}$ for all t . In particular, $\tilde{g}_1$ gives a map $L_\gamma: F_{\gamma(0)} \rightarrow F_{\gamma(1)}$. The association $\gamma \mapsto L_\gamma$ has the following basic properties:

- (a) If $\gamma \simeq \gamma' \text{ rel } \partial I$, then $L_\gamma \simeq L_{\gamma'}$. In particular the homotopy class of L_γ is independent of the choice of the lifting $\tilde{g}_t$ of g_t .
- (b) For a composition of paths $\gamma\gamma'$, $L_{\gamma\gamma'}$ is homotopic to the composition $L_{\gamma'}L_\gamma$.

From these statements it follows that L_γ is a homotopy equivalence with homotopy inverse $L_{\bar{\gamma}}$, where $\bar{\gamma}$ is the inverse path of γ .

Before proving (a), note that a fibration has the homotopy lifting property for pairs $(X \times I, X \times \partial I)$ since the pairs $(I \times I, I \times \{0\} \cup \partial I \times I)$ and $(I \times I, I \times \{0\})$ are homeomorphic, hence the same is true after taking products with X .

To prove (a), let $\gamma(s, t)$ be a homotopy from $\gamma(t)$ to $\gamma'(t)$, $(s, t) \in I \times I$. This determines a family $g_{st}: F_{\gamma(0)} \rightarrow B$ with $g_{st}(F_{\gamma(0)}) = \gamma(s, t)$. Let $\tilde{g}_{0,t}$ and $\tilde{g}_{1,t}$ be lifts defining L_γ and $L_{\gamma'}$, and let $\tilde{g}_{s,0}$ be the inclusion $F_{\gamma(0)} \hookrightarrow E$ for all s . Using the homotopy lifting property for the pair $(F_{\gamma(0)} \times I, F_{\gamma(0)} \times \partial I)$, we can extend these lifts to lifts $\tilde{g}_{st}$ for $(s, t) \in I \times I$. Restricting to $t = 1$ then gives a homotopy $L_\gamma \simeq L_{\gamma'}$.

Property (b) holds since for lifts $\tilde{g}_t$ and $\tilde{g}'_t$ defining L_γ and $L_{\gamma'}$ we obtain a lift defining $L_{\gamma\gamma'}$ by taking $\tilde{g}_{2t}$ for $0 \leq t \leq 1/2$ and $\tilde{g}'_{2t-1}L_{\gamma'}$ for $1/2 \leq t \leq 1$. $\square$

One may ask whether fibrations satisfy a homotopy analog of the local triviality property of fiber bundles. Observe first that for a fibration $p: E \rightarrow B$, the restriction

$p: p^{-1}(A) \rightarrow A$ is a fibration for any subspace $A \subset B$. So we can ask whether every point of B has a neighborhood U for which the fibration $p^{-1}(U) \rightarrow U$ is equivalent in some homotopy-theoretic sense to a projection $U \times F \rightarrow U$. The natural notion of equivalence for fibrations is defined in the following way. Given fibrations $p_1: E_1 \rightarrow B$ and $p_2: E_2 \rightarrow B$, a map $f: E_1 \rightarrow E_2$ is called **fiber-preserving** if $p_1 = p_2 f$, or in other words, $f(p_1^{-1}(b)) \subset p_2^{-1}(b)$ for all $b \in B$. A fiber-preserving map $f: E_1 \rightarrow E_2$ is a **fiber homotopy equivalence** if there is a fiber-preserving map $g: E_2 \rightarrow E_1$ such that both compositions fg and gf are homotopic to the identity through fiber-preserving maps. A fiber homotopy equivalence can be thought of as a family of homotopy equivalences between corresponding fibers of E_1 and E_2 . An interesting fact is that a fiber-preserving map that is a homotopy equivalence is a fiber homotopy equivalence; this is an exercise for §4.H.

We will show that a fibration $p: E \rightarrow B$ is locally fiber-homotopically trivial in the sense described above if B is locally contractible. In order to do this we first digress to introduce another basic concept.

Given a fibration $p: E \rightarrow B$ and a map $f: A \rightarrow B$, there is a **pullback** or **induced** fibration $f^*(E) \rightarrow A$ obtained by setting $f^*(E) = \{(a, e) \in A \times E \mid f(a) = p(e)\}$, with the projections of $f^*(E)$ onto A and E giving a commutative diagram as shown at the right. The homotopy lifting property holds for $f^*(E) \rightarrow A$ since a homotopy $g_t: X \rightarrow A$ gives the first coordinate of a lift $\tilde{g}_t: X \rightarrow f^*(E)$, the second coordinate being a lifting to E of the composed homotopy fg_t .

$$\begin{array}{ccc} f^*(E) & \longrightarrow & E \\ \downarrow & & \downarrow p \\ A & \xrightarrow{f} & B \end{array}$$

Proposition 4.62. *Given a fibration $p: E \rightarrow B$ and a homotopy $f_t: A \rightarrow B$, the pullback fibrations $f_0^*(E) \rightarrow A$ and $f_1^*(E) \rightarrow A$ are fiber homotopy equivalent.*

Proof: Let $F: A \times I \rightarrow B$ be the homotopy f_t . The fibration $F^*(E) \rightarrow A \times I$ contains $f_0^*(E)$ and $f_1^*(E)$ over $A \times \{0\}$ and $A \times \{1\}$. So it suffices to prove the following: For a fibration $p: E \rightarrow B \times I$, the restricted fibrations $E_s = p^{-1}(B \times \{s\}) \rightarrow B$ are all fiber homotopy equivalent for $s \in [0, 1]$.

To prove this assertion the idea is to imitate the construction of the homotopy equivalences L_y in the proof of Proposition 4.61. A path $y: [0, 1] \rightarrow I$ gives rise to a fiber-preserving map $L_y: E_{y(0)} \rightarrow E_{y(1)}$ by lifting the homotopy $g_t: E_{y(0)} \rightarrow B \times I$, $g_t(x) = (p(x), y(t))$, starting with the inclusion $E_{y(0)} \hookrightarrow E$. As before, one shows the two basic properties (a) and (b), noting that in (a) the homotopy $L_y \simeq L_{y'}$ is fiber-preserving since it is obtained by lifting a homotopy $h_t: E_{y(0)} \times [0, 1] \rightarrow B \times I$ of the form $h_t(x, u) = (p(x), -)$. From (a) and (b) it follows that L_y is a fiber homotopy equivalence with inverse $L_{\bar{y}}$. $\square$

Corollary 4.63. *A fibration $E \rightarrow B$ over a contractible base B is fiber homotopy equivalent to a product fibration $B \times F \rightarrow B$.*

Proof: The pullback of E by the identity map $B \rightarrow B$ is E itself, while the pullback by a constant map $B \rightarrow B$ is a product $B \times F$. $\square$

Thus we see that if B is locally contractible then any fibration over B is locally fiber homotopy equivalent to a product fibration.

Pathspace Constructions

There is a simple but extremely useful way to turn arbitrary mappings into fibrations. Given a map $f: A \rightarrow B$, let E_f be the space of pairs (a, γ) where $a \in A$ and $\gamma: I \rightarrow B$ is a path in B with $\gamma(0) = f(a)$. We topologize E_f as a subspace of $A \times B^I$, where B^I is the space of mappings $I \rightarrow B$ with the compact-open topology; see the Appendix for the definition and basic properties of this topology, in particular Proposition A.14 which we will be using shortly.

Proposition 4.64. *The map $p: E_f \rightarrow B$, $p(a, \gamma) = \gamma(1)$, is a fibration.*

Proof: Continuity of p follows from (a) of Proposition A.14 in the Appendix which says that the evaluation map $B^I \times I \rightarrow B$, $(\gamma, s) \mapsto \gamma(s)$, is continuous.

To verify the fibration property, let a homotopy $g_t: X \rightarrow B$ and a lift $\tilde{g}_0: X \rightarrow E_f$ of g_0 be given. Write $\tilde{g}_0(x) = (h(x), \gamma_x)$ for $h: X \rightarrow A$ and $\gamma_x: I \rightarrow B$. Define a lift $\tilde{g}_t: X \rightarrow E_f$ by $\tilde{g}_t(x) = (h(x), \gamma_x \cdot g_{[0,t]}(x))$, the second coordinate being the path γ_x followed by the path traced out by $g_s(x)$ for $0 \leq s \leq t$. This composition of paths is defined since $g_0(x) = p\tilde{g}_0(x) = \gamma_x(1)$. To check that $\tilde{g}_t$ is a continuous homotopy we regard it as a map $X \times I \rightarrow E_f \subset A \times B^I$ and then apply (b) of Proposition A.14 which in the current context asserts that continuity of a map $X \times I \rightarrow A \times B^I$ is equivalent to continuity of the associated map $X \times I \times I \rightarrow A \times B$. $\square$

We can regard A as the subspace of E_f consisting of pairs (a, γ) with γ the constant path at $f(a)$, and E_f deformation retracts onto this subspace by restricting all the paths γ to shorter and shorter initial segments. The map $p: E_f \rightarrow B$ restricts to f on the subspace A , so we have factored an arbitrary map $f: A \rightarrow B$ as the composition $A \hookrightarrow E_f \rightarrow B$ of a homotopy equivalence and a fibration. We can also think of this construction as extending f to a fibration $E_f \rightarrow B$ by enlarging its domain to a homotopy equivalent space. The fiber F_f of $E_f \rightarrow B$ is called the **homotopy fiber** of f . It consists of all pairs (a, γ) with $a \in A$ and γ a path in B from $f(a)$ to a basepoint $b_0 \in B$.

If $f: A \rightarrow B$ is the inclusion of a subspace, then E_f is the space of paths in B starting at points of A . In this case a map $(I^{i+1}, \partial I^{i+1}, J^i) \rightarrow (B, A, x_0)$ is the same as a map $(I^i, \partial I^i) \rightarrow (F_f, \gamma_0)$ where γ_0 is the constant path at x_0 and F_f is the fiber of E_f over x_0 . This means that $\pi_{i+1}(B, A, x_0)$ can be identified with $\pi_i(F_f, \gamma_0)$, hence the long exact sequences of homotopy groups of the pair (B, A) and of the fibration $E_f \rightarrow B$ can be identified.

An important special case is when f is the inclusion of the basepoint b_0 into B . Then E_f is the space PB of paths in B starting at b_0 , and $p:PB \rightarrow B$ sends each path to its endpoint. The fiber $p^{-1}(b_0)$ is the loop space ΩB consisting of all loops in B based at b_0 . Since PB is contractible by progressively truncating paths, the long exact sequence of homotopy groups for the path fibration $PB \rightarrow B$ yields another proof that $\pi_n(X, x_0) \approx \pi_{n-1}(\Omega X, x_0)$ for all n .

As we mentioned in the discussion of loop spaces earlier in this section, it is a theorem of [Milnor 1959] that the loop space of a CW complex is homotopy equivalent to a CW complex. Milnor's theorem is actually quite a bit more general than this, and implies in particular that the homotopy fiber of an arbitrary map between CW complexes has the homotopy type of a CW complex. One can usually avoid quoting these results by using CW approximations, though it is reassuring to know they are available if needed, or if one does not want to bother with CW approximations.

If the fibration construction $f \mapsto E_f$ is applied to a map $p:E \rightarrow B$ that is already a fibration, one might expect the resulting fibration $E_p \rightarrow B$ to be closely related to the original fibration $E \rightarrow B$. This is indeed the case:

Proposition 4.65. *If $p:E \rightarrow B$ is a fibration, then the inclusion $E \hookrightarrow E_p$ is a fiber homotopy equivalence. In particular, the homotopy fibers of p are homotopy equivalent to the actual fibers.*

Proof: We apply the homotopy lifting property to the homotopy $g_t:E_p \rightarrow B$, $g_t(e, \gamma) = \gamma(t)$, with initial lift $\tilde{g}_0:E_p \rightarrow E$, $\tilde{g}_0(e, \gamma) = e$. The lifting $\tilde{g}_t:E_p \rightarrow E$ is then the first coordinate of a homotopy $h_t:E_p \rightarrow E_p$ whose second coordinate is the restriction of the paths γ to the interval $[t, 1]$. Since the endpoints of the paths γ are unchanged, h_t is fiber-preserving. We have $h_0 = \mathbb{1}$, $h_1(E_p) \subset E$, and $h_t(E) \subset E$ for all t . If we let i denote the inclusion $E \hookrightarrow E_p$, then $ih_1 \simeq \mathbb{1}$ via h_t and $h_1 i \simeq \mathbb{1}$ via $h_t|_E$, so i is a fiber homotopy equivalence. $\square$

We have seen that loop spaces occur as fibers of fibrations $PB \rightarrow B$ with contractible total space PB . Here is something of a converse:

Proposition 4.66. *If $F \rightarrow E \rightarrow B$ is a fibration or fiber bundle with E contractible, then there is a weak homotopy equivalence $F \rightarrow \Omega B$.*

Proof: If we compose a contraction of E with the projection $p:E \rightarrow B$ then we have for each point $x \in E$ a path γ_x in B from $p(x)$ to a basepoint $b_0 = p(x_0)$, where x_0 is the point to which E contracts. This yields a map $E \rightarrow PB$, $x \mapsto \overline{\gamma}_x$, whose composition with the fibration $PB \rightarrow B$ is p . By restriction this gives a map $F \rightarrow \Omega B$ where $F = p^{-1}(b_0)$, and the long exact sequence of homotopy groups for $F \rightarrow E \rightarrow B$ maps to the long exact sequence for $\Omega B \rightarrow PB \rightarrow B$. Since E and PB are contractible, the five-lemma implies that the map $F \rightarrow \Omega B$ is a weak homotopy equivalence. $\square$

Examples arising from fiber bundles constructed earlier in the chapter are $O(n) \simeq \Omega G_n(\mathbb{R}^\infty)$, $U(n) \simeq \Omega G_n(\mathbb{C}^\infty)$, and $Sp(n) \simeq \Omega G_n(\mathbb{H}^\infty)$. In particular, taking $n = 1$ in the latter two examples, we have $S^1 \simeq \Omega \mathbb{C}P^\infty$ and $S^3 \simeq \Omega \mathbb{H}P^\infty$. Note that in all these examples it is a topological group that is homotopy equivalent to a loop space. In [Milnor 1956] this is shown to hold in general: For each topological group G there is a fiber bundle $G \rightarrow EG \rightarrow BG$ with EG contractible, hence by the proposition there is a weak equivalence $G \simeq \Omega BG$. There is also a converse statement: The loop space of a CW complex is homotopy equivalent to a topological group.

The relationship between X and ΩX has been much studied, particularly the case that ΩX has the homotopy type of a finite CW complex, which is of special interest because of the examples of the classical Lie groups such as $O(n)$, $U(n)$, and $Sp(n)$. See [Kane 1988] for an introduction to this subject.

It is interesting to see what happens when the process of forming homotopy fibers is iterated. Given a fibration $p: E \rightarrow B$ with fiber $F = p^{-1}(b_0)$, we know that the inclusion of F into the homotopy fiber F_p is a homotopy equivalence. Recall that F_p consists of pairs (e, γ) with $e \in E$ and γ a path in B from $p(e)$ to b_0 . The inclusion $F \hookrightarrow E$ extends to a map $i: F_p \rightarrow E$, $i(e, \gamma) = e$, and this map is obviously a fibration. In fact it is the pullback via p of the path fibration $PB \rightarrow B$. This allows us to iterate, taking the homotopy fiber F_i with its map to F_p , and so on, as in the first row of the following diagram:

$$\begin{array}{ccccccccc} \cdots & \longrightarrow & F_j & \longrightarrow & F_i & \xrightarrow{j} & F_p & \xrightarrow{i} & E & \xrightarrow{p} & B \\ & & \simeq \downarrow & & \simeq \downarrow & & \simeq \uparrow & & \parallel & & \parallel \\ \cdots & \longrightarrow & \Omega E & \xrightarrow{\Omega p} & \Omega B & \longrightarrow & F & \longrightarrow & E & \xrightarrow{p} & B \end{array}$$

The actual fiber of i over a point $e_0 \in p^{-1}(b_0)$ consists of pairs (e_0, γ) with γ a loop in B at the basepoint b_0 , so this fiber is just ΩB , and the inclusion $\Omega B \hookrightarrow F_i$ is a homotopy equivalence. In the second row of the diagram the map $\Omega B \rightarrow F$ is the composition $\Omega B \hookrightarrow F_i \rightarrow F_p \rightarrow F$ where the last map is a homotopy inverse to the inclusion $F \hookrightarrow F_p$, so the square in the diagram containing these maps commutes up to homotopy. The homotopy fiber F_i consists of pairs (γ, η) where η is a path in E ending at e_0 and γ is a path in B from $p(\eta(0))$ to b_0 . A homotopy inverse to the inclusion $\Omega B \hookrightarrow F_i$ is the retraction $F_i \rightarrow \Omega B$ sending (γ, η) to the loop obtained by composing the inverse path of $p\eta$ with γ . These constructions can now be iterated indefinitely.

Thus we produce a sequence

$$\cdots \rightarrow \Omega^2 B \rightarrow \Omega F \rightarrow \Omega E \rightarrow \Omega B \rightarrow F \rightarrow E \rightarrow B$$

where any two consecutive maps form a fibration, up to homotopy equivalence, and all the maps to the left of ΩB are obtained by applying the functor Ω to the later maps. The long exact sequence of homotopy groups for any fibration in the sequence coincides with the long exact sequence for $F \rightarrow E \rightarrow B$, as the reader can check.

Postnikov Towers

A Postnikov tower for a path-connected space X is a commutative diagram as at the right, such that:

- (1) The map $X \rightarrow X_n$ induces an isomorphism on π_i for $i \leq n$.
- (2) $\pi_i(X_n) = 0$ for $i > n$.

As we saw in Example 4.17, every connected CW complex X has a Postnikov tower, and this is unique up to homotopy equivalence by Corollary 4.19.

If we convert the map $X_n \rightarrow X_{n-1}$ into a fibration, its fiber F_n is a $K(\pi_n X, n)$, as is apparent from a brief inspection of the long exact sequence of homotopy groups for the fibration:

$$\pi_{i+1}(X_n) \rightarrow \pi_{i+1}(X_{n-1}) \rightarrow \pi_i(F_n) \rightarrow \pi_i(X_n) \rightarrow \pi_i(X_{n-1})$$

We can replace each map $X_n \rightarrow X_{n-1}$ by a fibration $X'_n \rightarrow X'_{n-1}$ in succession, starting with $X_2 \rightarrow X_1$ and working upward. For the inductive step we convert the composition $X_n \rightarrow X_{n-1} \hookrightarrow X'_{n-1}$ into a fibration $X'_n \rightarrow X'_{n-1}$ fitting into the commutative diagram at the right. Thus we obtain a Postnikov tower satisfying also the condition

- (3) The map $X_n \rightarrow X_{n-1}$ is a fibration with fiber a $K(\pi_n X, n)$.

To the extent that fibrations can be regarded as twisted products, up to homotopy equivalence, the spaces X_n in a Postnikov tower for X can be thought of as twisted products of Eilenberg-MacLane spaces $K(\pi_n(X), n)$.

For many purposes, a CW complex X can be replaced by one of the stages X_n in a Postnikov tower for X , for example if one is interested in homotopy or homology groups in only a finite range of dimensions. However, to determine the full homotopy type of X from its Postnikov tower, some sort of limit process is needed. Let us investigate this question in somewhat greater generality.

Given a sequence of maps $\cdots \rightarrow X_2 \rightarrow X_1$, define their **inverse limit** $\varprojlim X_n$ to be the subspace of the product $\prod_n X_n$ consisting of sequences of points $x_n \in X_n$ with x_n mapping to x_{n-1} under the map $X_n \rightarrow X_{n-1}$. The corresponding algebraic notion is the inverse limit $\varprojlim X_n$ of a sequence of group homomorphisms $\cdots \rightarrow G_2 \rightarrow G_1$, which is the subgroup of $\prod_n G_n$ consisting of sequences of elements $g_n \in G_n$ with g_n mapping to g_{n-1} under the homomorphism $G_n \rightarrow G_{n-1}$.

Proposition 4.67. *For an arbitrary sequence of fibrations $\cdots \rightarrow X_2 \rightarrow X_1$ the natural map $\lambda: \pi_i(\varprojlim X_n) \rightarrow \varprojlim \pi_i(X_n)$ is surjective, and λ is injective if the maps $\pi_{i+1}(X_n) \rightarrow \pi_{i+1}(X_{n-1})$ are surjective for n sufficiently large.*

Proof: Represent an element of $\varprojlim \pi_i(X_n)$ by maps $f_n: (S^i, s_0) \rightarrow (X_n, x_n)$. Since the projection $p_n: X_n \rightarrow X_{n-1}$ takes $[f_n]$ to $[f_{n-1}]$, by applying the homotopy lifting

